

Political Ideology Classification from Congressional Tweets

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1. Introduction

In recent years, social media platforms such as Twitter have become important channels for political communication. Members of Congress frequently use Twitter to express political opinions, respond to public events, and communicate with voters. Because of this, political tweets provide a valuable source of text data that can be analyzed with machine learning techniques.

The main goal of this project is to examine whether political ideology can be detected from the textual content of tweets. Specifically, the project explores whether machine learning models can classify tweets based on political party affiliation. This question is interesting because it helps us understand whether ideological differences appear clearly in language patterns used by politicians.

If this problem can be solved effectively, it could have several impacts. First, it may help researchers analyze political discourse at large scale. Second, it could help identify ideological patterns in communication. Third, it may provide insights into how political narratives are constructed online.

In this project, both supervised and unsupervised machine learning methods are applied. Supervised learning models attempt to classify tweets based on party labels, while unsupervised learning methods explore whether tweets naturally cluster into meaningful groups.

The supervised learning models tested in this project include Logistic Regression, Linear Support Vector Machines, Multinomial Naive Bayes, and Random Forest. For unsupervised learning, clustering methods were used to examine underlying structure in the tweet corpus.

The main finding from supervised learning is that linear models perform best, especially Linear SVM with an F1 score around 0.85. This suggests that lexical signals captured by TF-IDF features are strong indicators of ideological differences. In contrast, tree-based models performed worse on sparse text features.

The unsupervised results show that tweets can be grouped into clusters representing different topics and communication styles. However, these clusters do not always correspond exactly to party labels.

2. Related Work

Text classification has been widely studied in natural language processing and political science. Several previous studies have used machine learning methods to analyze political communication on social media.

One line of work focuses on ideological classification of political texts. For example, previous research has used supervised machine learning models to classify congressional speeches or political manifestos. These studies often rely on bag-of-words or TF-IDF features combined with linear classifiers.

Another related area is sentiment and discourse analysis in political tweets. Some studies examine how politicians communicate policy messages or respond to political events on Twitter. These studies often use topic modeling or clustering methods to identify common themes in political communication.

Compared to previous work, this project combines both supervised and unsupervised learning in a single framework. Instead of focusing only on classification accuracy, the project also explores the underlying structure of the tweet dataset using clustering analysis. This allows a better understanding of how political language is organized.

This project also extends earlier coursework by building a full machine learning pipeline including feature engineering, model comparison, cross-validation, and cluster analysis.

3. Data Sources

The dataset used in this project consists of tweets posted by members of the United States Congress. The data includes tweets from both Democratic and Republican politicians.

The dataset was provided as a CSV file containing tweet text and associated metadata. The most important variables in the dataset include:

- Tweet text
- Party affiliation label
- Tweet ID and metadata

The dataset contains several thousand tweets collected from congressional Twitter accounts. These tweets represent a mixture of policy discussion, campaign messaging, and responses to current events.

Because the raw data consists of unstructured text, several preprocessing steps were required before machine learning models could be applied. These steps include tokenization, stopwords removal, and feature transformation using TF-IDF vectorization.

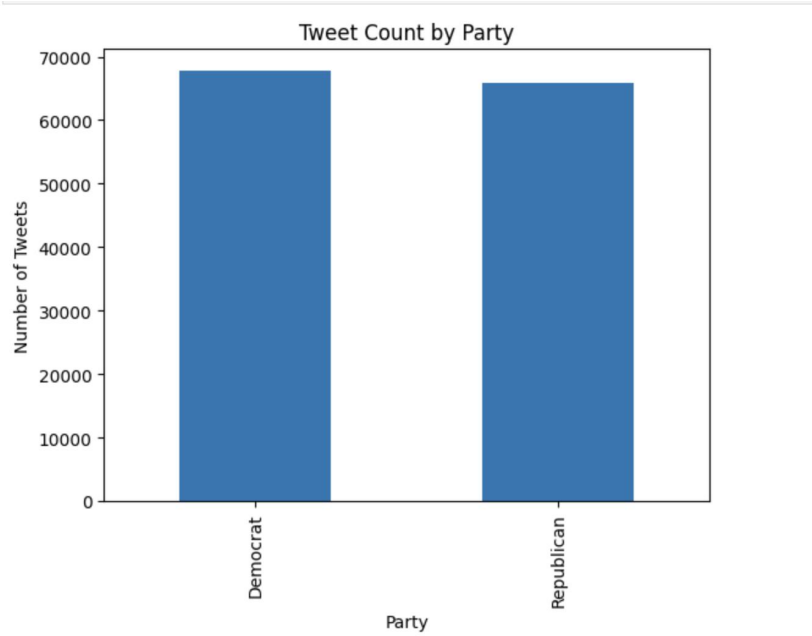


Figure 1. Number of tweets by party in the cleaned congressional tweet dataset

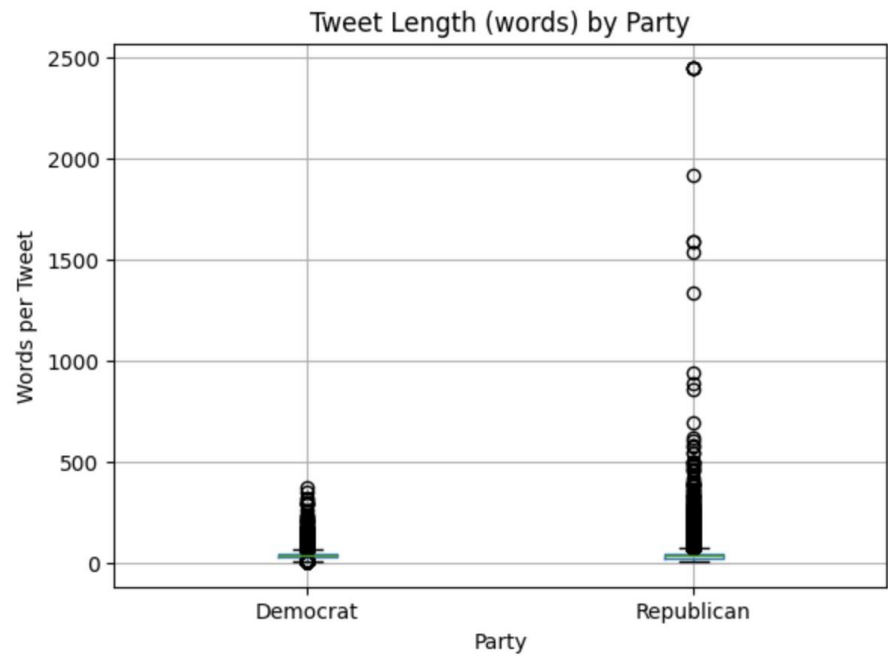


Figure 2. Distribution of tweet length in words by party

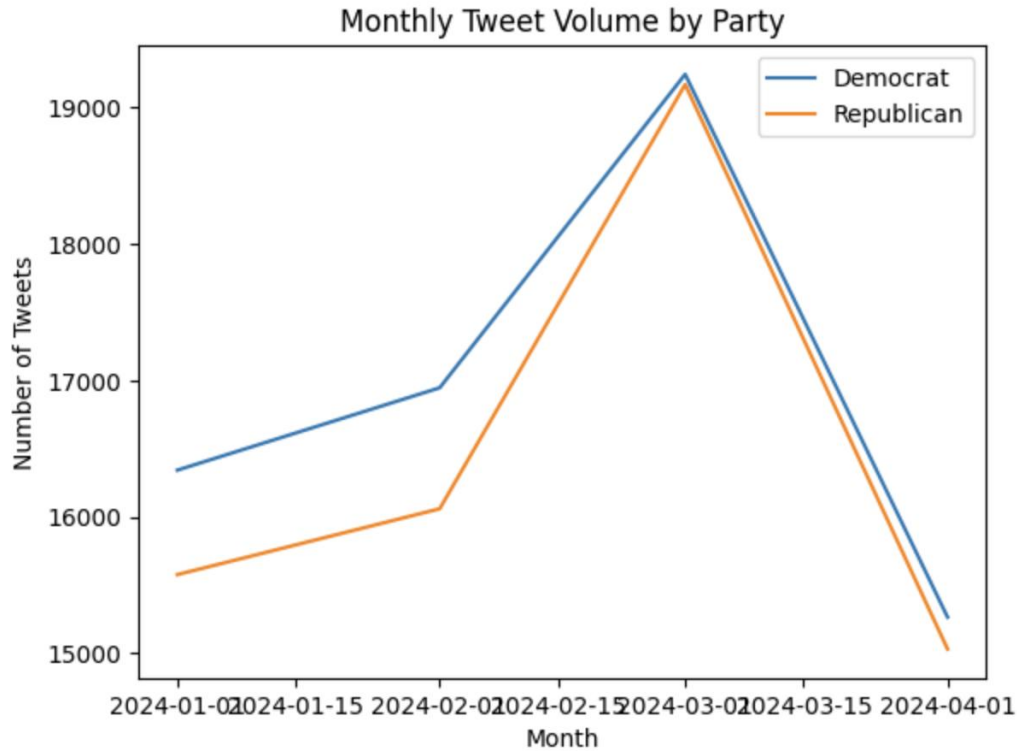


Figure 3. Monthly tweet volume by party during the observed time period.

4. Feature Engineering

The raw tweet text cannot be directly used by machine learning models. Therefore several preprocessing and feature engineering steps were applied.

First, basic text preprocessing was performed. This includes converting text to lowercase, removing punctuation, and eliminating common English stopwords. These steps help reduce noise in the dataset.

Next, TF-IDF vectorization was used to convert text into numerical features. TF-IDF stands for Term Frequency–Inverse Document Frequency. This method assigns higher weights to words that are important within a document but less common across the entire corpus.

In this project, both unigram and bigram features were used. This allows the model to capture not only individual words but also short phrases that may carry political meaning.

Because the TF-IDF representation produces very high dimensional feature vectors, dimensionality reduction was also applied using TruncatedSVD. This step helps reduce sparsity and improves computational efficiency for certain models.

The final feature representation consists of TF-IDF weighted word features, optionally projected into lower dimensional space using SVD.

5. Supervised Learning Methods

Several supervised machine learning models were used in this project. The goal of these models is to classify tweets according to party affiliation.

The first model is Logistic Regression. Logistic regression is a widely used linear classifier that performs well for high-dimensional text data. It is also relatively interpretable because feature weights can indicate important words.

The second model is Linear Support Vector Machines (Linear SVM). SVMs are known to perform well in text classification tasks due to their ability to handle high-dimensional sparse data.

The third model is Multinomial Naive Bayes. This model is commonly used in text classification because it models word frequencies using probabilistic assumptions.

The fourth model is Random Forest, which represents a tree-based model family. Random Forest can capture nonlinear relationships but sometimes struggles with sparse text features.

To evaluate these models fairly, five-fold cross-validation was used. The evaluation metrics include accuracy and F1 score.

6. Supervised Evaluation

The results show clear differences between model families. Accuracy and macro F1 score were used as the primary evaluation metrics because the dataset is relatively balanced across parties. Linear SVM achieved the highest performance with an average F1 score around 0.85. Logistic Regression produced slightly lower but still strong results. Multinomial Naive Bayes performed moderately well but slightly below the linear models.

Random Forest performed significantly worse than the linear models. One possible reason is that tree-based models do not handle sparse high-dimensional TF-IDF features as efficiently as linear classifiers.

Overall, these results suggest that linear classifiers remain strong baselines for text classification tasks.

[21:04:03] END: CV RF | 168.98s

	model	acc_mean	acc_std	f1_mean	f1_std
0	LinearSVM	0.850137	0.002588	0.850083	0.002603
1	LogReg	0.842612	0.002952	0.842486	0.002950
2	MultinomialNB	0.826100	0.002536	0.825751	0.002562
3	TFIDF+SVD+RandomForest	0.769050	0.001089	0.767596	0.001149

Table 1. Five-fold cross-validation performance of supervised models.

As shown in Table 1, LinearSVM achieved the best overall performance, followed closely by Logistic Regression. Random Forest performed substantially worse than the linear models.

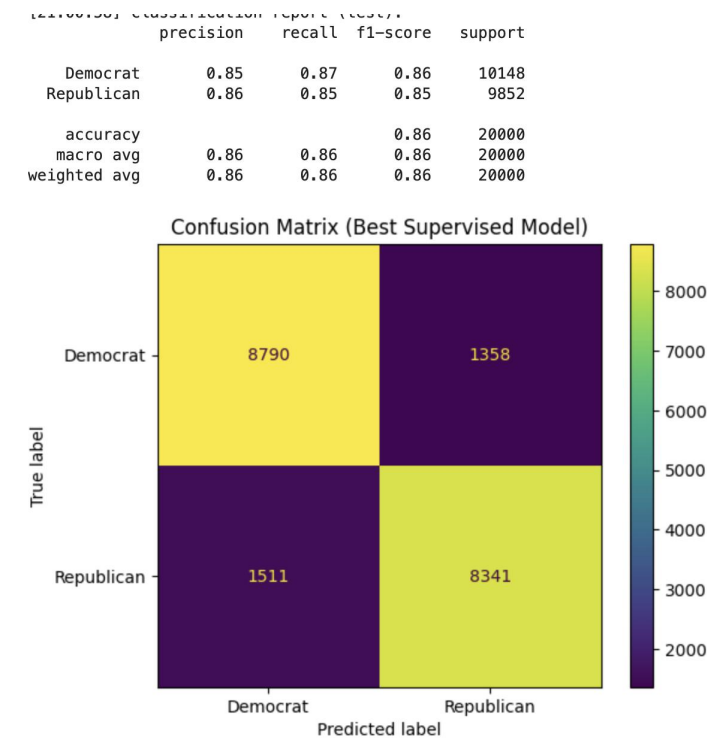


Figure 4. Confusion matrix for the best supervised model on the held-out test set.

Figure 4 shows that the best model performs relatively well on both classes, with balanced error rates for Democrats and Republicans.

Feature importance analysis also shows that certain political keywords strongly influence classification. Words related to policy issues such as immigration, healthcare, and taxation appear frequently among the most predictive features.

[21:01:02] ablation table:

	ngram	f1_mean	f1_std
0	(1, 1)	0.834377	0.002807
1	(1, 2)	0.851167	0.002891

Table 3. Ablation results for unigram versus unigram-plus-bigram TF-IDF features.

Table 3 shows that adding bigrams improves performance, suggesting that short phrases carry useful partisan signals beyond single words.

Sensitivity analysis indicates that model performance remains relatively stable across different TF-IDF parameter settings such as n-gram range and minimum document frequency.

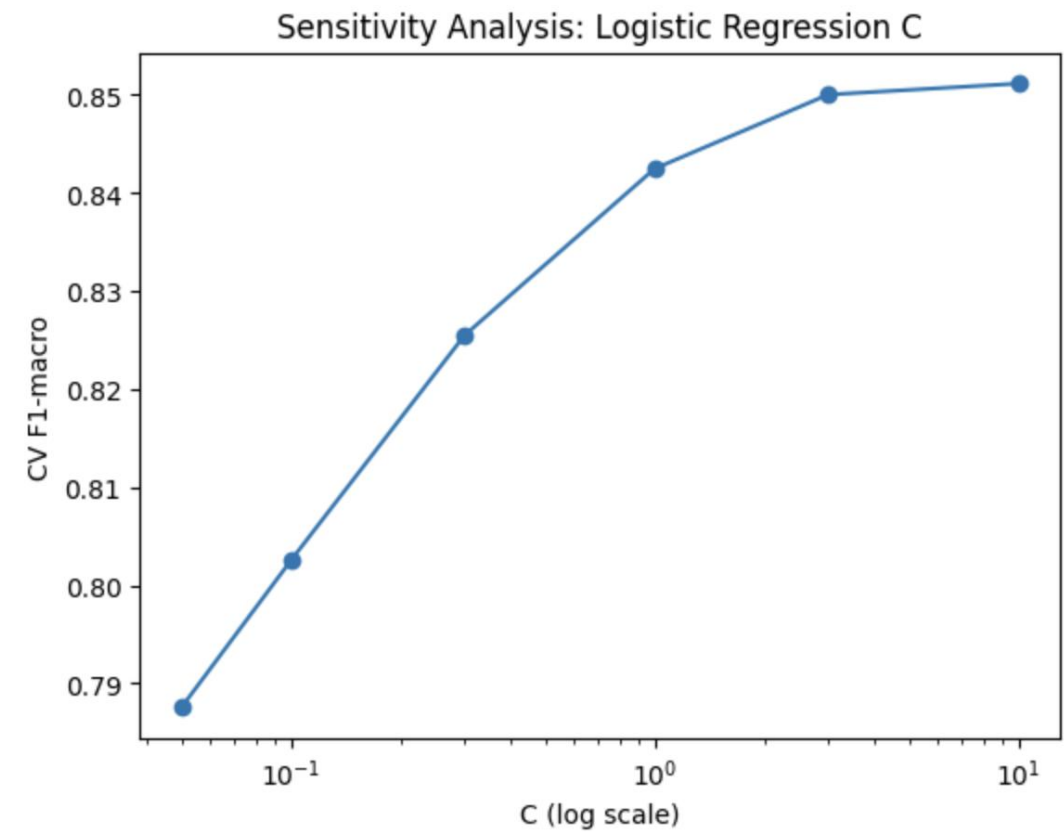


Figure 5. Sensitivity of Logistic Regression performance to different values of C.

	C	f1_mean	f1_std
0	0.05	0.787553	0.001603
1	0.10	0.802539	0.001907
2	0.30	0.825486	0.001686
3	1.00	0.842486	0.002950
4	3.00	0.850040	0.001585
5	10.00	0.851167	0.002891

Table 4. Cross-validated macro-F1 across different Logistic Regression C values.

Figure 5 and Table 4 show that performance increases as regularization becomes weaker, but the gains become very small after C reaches around 3.

7. Failure Analysis

Although the supervised models perform well overall, several types of classification errors were observed.

[21:01:05] misclassified count = 2869 out of 20000

	true	pred	text
0	Republican	Democrat	as the situation worsens at professor massad h...
1	Republican	Democrat	what an absolute tragedy my prayers are with t...
2	Democrat	Republican	i am extremely concerned about the invasion of...
3	Republican	Democrat	rt soybean growers are concerned about the rec...
4	Republican	Democrat	i had the privilege of speaking with members o...
5	Democrat	Republican	stay tuned for more information on the congres...
6	Democrat	Republican	good news this week the house unanimously pass...
7	Republican	Democrat	thanks to the for stopping by my office to dis...
8	Republican	Democrat	american families can t afford to wait for us ...
9	Democrat	Republican	passenger safety is non negotiable these are c...

Table 5. Example misclassified tweets from the best supervised model.

Table 5 presents representative misclassified cases. Many of them involve neutral institutional language, bipartisan policy language, or short tweets with limited contextual cues.

The first category of failure involves tweets discussing bipartisan issues. These tweets often use neutral policy language that does not strongly signal party affiliation.

The second type of failure occurs with very short tweets. Short tweets sometimes contain limited contextual information, making it difficult for the model to identify ideological signals.

A third type of error occurs when tweets contain sarcasm or quotes from opposing politicians. In these cases, the language may resemble the opposing party's rhetoric, which confuses the classifier.

These failure cases suggest that incorporating additional context, such as user metadata or conversation threads, could improve classification accuracy.

8. Unsupervised Learning

In addition to supervised classification, unsupervised learning methods were used to explore the structure of the tweet dataset.

KMeans clustering was applied to the TF-IDF feature representation. Before clustering, dimensionality reduction using TruncatedSVD was performed to project the data into a lower-dimensional semantic space.

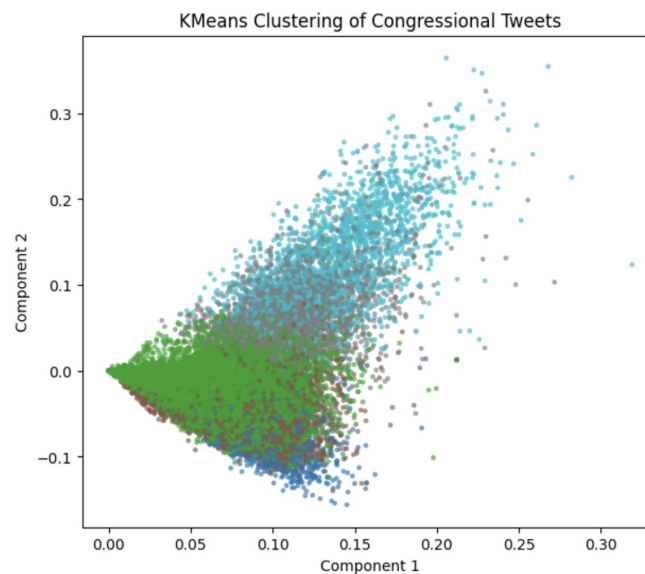


Figure 6. KMeans clustering of congressional tweets

cluster		top_words
0	0	aanhpi, abandoning, abandoned, abbey, abdullah...
1	1	aanhpi, abandon, ability control, abetting, ab...
2	2	aanhpi, abhorrent, abdication, abby, aaron, ab...
3	3	aanhpi, aapi, aanhpi communities, abbey, Abdul...
4	4	aanhpi, aanhpi communities, aaron, abandoned, ...

Table 6. Cluster Keywords Table

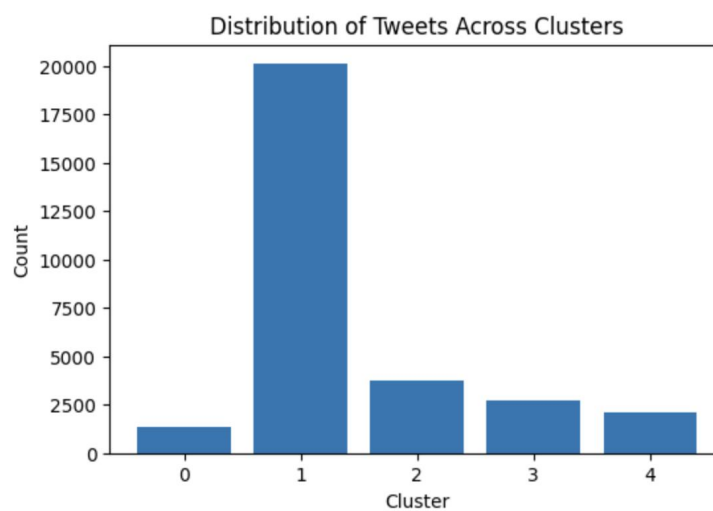


Figure 7. Distribution of tweets across the five clusters identified by KMeans.

[22103140] Running K		
	k	inertia
0	3	1981.758537
1	4	1933.396972
2	5	1894.044851
3	6	1852.448634
4	7	1816.447320
5	8	1783.626253

Table 8 Inertia across different values of k

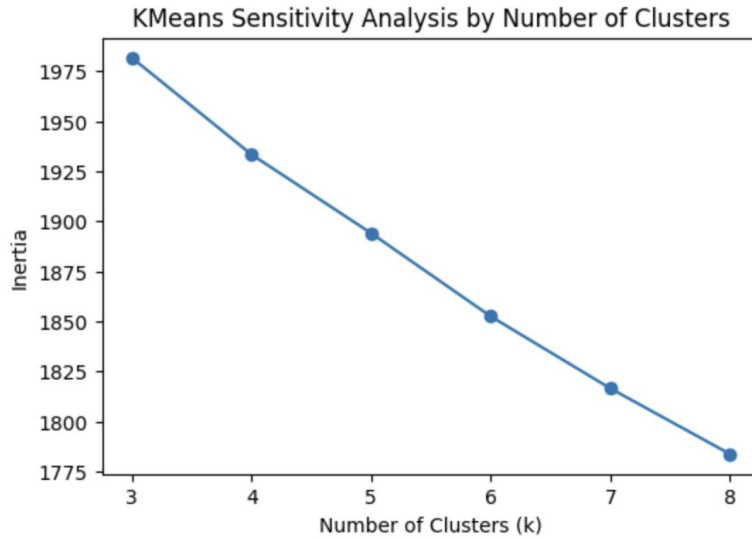


Figure 8. Sensitivity of KMeans to the choice of k

To explore the internal structure of the tweet corpus, I applied unsupervised learning using TF-IDF features, TruncatedSVD, and KMeans clustering. I used a random sample of 30,000 tweets for the unsupervised analysis in order to keep computation manageable. After TF-IDF vectorization, I reduced the feature space to 50 dimensions using TruncatedSVD and then applied KMeans with $k=5$. Figure 6 shows the clustering structure in a 2D projection, while Figure 7 and Table 6 summarize the cluster size distribution. Table 7 presents representative keywords for each cluster. In addition, I tested the sensitivity of the clustering results to different values of k , and Table 8 / Figure 8 show that inertia decreases steadily as the number of clusters increases. These results suggest that the tweet dataset contains some internal thematic structure, although some clusters are easier to interpret than others.

9. Discussion

From the supervised learning experiments, one important lesson is that simple linear models can perform very well for political text classification. Despite their simplicity, models such as Linear SVM can achieve high accuracy when combined with strong feature representations like TF-IDF.

Another interesting observation is that more complex models such as Random Forest do not necessarily perform better. In fact, they may perform worse when applied to sparse text features.

From the unsupervised learning analysis, clustering reveals that political communication contains several thematic groups. However, these clusters do not perfectly align with party labels. This suggests that politicians from different parties sometimes discuss similar topics.

One challenge encountered in this project was the high dimensionality of text data. Dimensionality reduction techniques such as TruncatedSVD helped address this issue.

With more time and resources, future work could explore transformer-based language models such as BERT or other contextual embeddings to capture deeper semantic information.

10. Ethical Considerations

There are several ethical issues related to automated political text analysis.

First, automated classification of political ideology could potentially be misused for political surveillance or censorship. It is important that such tools are used responsibly and transparently.

Second, machine learning models may reinforce existing political biases if they are trained on biased datasets. Researchers should carefully consider how training data is collected and labeled.

Finally, political discourse is complex and context-dependent. Machine learning models should not be interpreted as definitive judgments about political ideology.

11. Statement of Work

Wuke Zhang implemented the data preprocessing pipeline, feature engineering, supervised learning models, and clustering analysis. The author also conducted evaluation and prepared the final project report.

12. References

Pedregosa et al. (2011). Scikit-learn: Machine Learning in Python.

Jurafsky & Martin (2023). Speech and Language Processing.

Scikit-learn Documentation. <https://scikit-learn.org>

Appendix A. Reproducibility Notes

Python 3.14

pandas 2.3.3

scikit-learn 1.7.2

TF-IDF settings

sample sizes: supervised 100000, unsupervised 30000